

Lecture #9: Scaling LLM - Basics [Kaplan + 2020 paper]

for a scenario: 10,000 H100 GPUs to build a open source LLM within one month.

How? ①: put together your infra team and distributed training framework (A2)

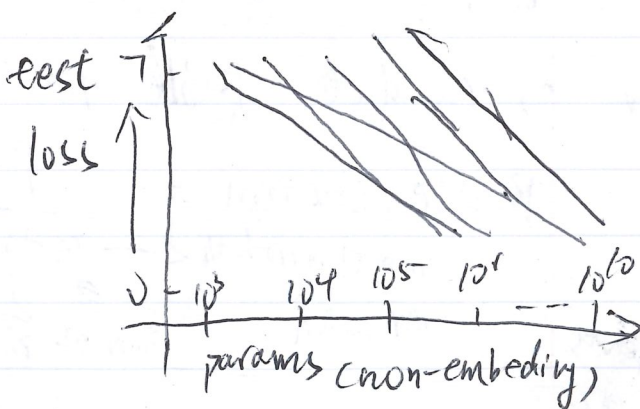
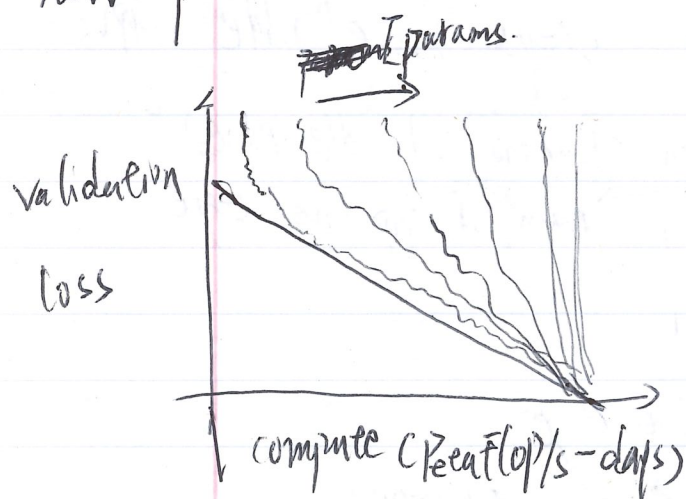
②: put together a great pre-training dataset (A4)

③: Run a big model (but which one??) ← we are here.

Today: simple, predictive 'laws' for behaviors of LLMs

Traditional: tune hyperparameters on big models

new optimism: tune on small models, extrapolate to large ones
scale-up



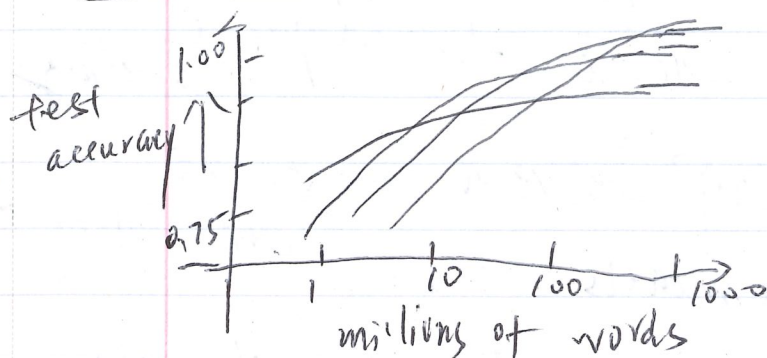
Part ①: scaling laws, history and background

⇒ Data scaling as empirical sample complexities

⇒ initial forays into understanding neural scaling w/ data

Earliest scaling (law) paper: Learning curve: Asymptotic values and Rate of convergence

Learning curves for confusion set: data scaling



↓ emergence
 ↓ scaling by compute
 ↓ accuracy = speed x resources

● Part #2: Neural (LLM) scaling behaviors

D: (1) Data vs performance: Are there simple rules that determine how data affects performance?

(2) Data vs model size: Do we train on more data or bigger models?

(3) Hyper-parameters vs performance: How should we set hyperparameters on the big model?

